Investigation on Properties of Abelian Sandpile Model on Peano Fractal Boundaries

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1 Background: Abelian Sandpile Model and Peano Curve Construction

In this section, we introduce the fundamental mechanisms of the Abelian Sandpile Model (ASM) and the recursive tiling construction of the space-filling Peano curve that serves as the boundary of our grid.

1.1 The Abelian Sandpile Model (ASM) Mechanism

The Abelian Sandpile Model is a discrete dynamical system defined on a graph $G = (V, E)$. In our system, the vertex set V is partitioned into two disjoint subsets: the active (bulk) vertices V_{active} and the sink vertices V_{sink}.

A sandpile configuration is a function $h : V_{\text{active}} \to \mathbb{Z}_{\geq 0}$ representing the number of sand grains at each active vertex. Since our underlying graph is a 2D square grid, each vertex has a degree of 4. A vertex $x \in V_{\text{active}}$ is defined to be stable if its height satisfies $h(x) < 4$, and unstable if $h(x) \geq 4$. An unstable vertex x can topple, losing 4 grains of sand and distributing exactly 1 grain to each of its four nearest neighbors:

$$\begin{aligned} h(x) &\to h(x) - 4 \\ h(y) &\to h(y) + 1 \quad \text{for } y \sim x \end{aligned}$$

Grains distributed to any sink vertices $y \in V_{\text{sink}}$ are permanently absorbed and removed from the system, acting as open boundary conditions. The stabilization process of toppling unstable vertices continues sequentially until the system reaches a unique stable configuration. Because the final state is independent of the order of topplings, the model is termed "Abelian".

The identity element E of this group is defined as the unique configuration that acts as the algebraic identity. That is, for any recurrent configuration u:

$$u \oplus E = (u + E)^{\circ} = u$$

where $(\cdot)^{\circ}$ represents the stabilization operator. Although E is formally defined by this group property, it can be computed using Dhar's burning algorithm (as detailed in Section 3), which reveals a non-trivial, highly intricate fractal-like pattern.

1.2 Recursive Peano Curve Construction

The boundary and sink set V_{sink} of our grid graph is defined by a space-filling Peano curve. The geometry of the curve is generated recursively using a set of two base tile types: the "upper" tile (Green, denoted by G) and the "lower" tile (Blue, denoted by B).

As illustrated in Figure 1, at Order 0, the base geometries of the tiles are defined by a black path winding through the square domain. The black path represents the curve itself, while the remaining colored background represents the partitioned bulk domains.

For any recursive order $k \geq 1$, each tile of size $3^k \times 3^k$ is constructed by subdividing it into a 3×3 grid of sub-tiles of order $k-1$. These sub-tiles are selected from $\{G, B\}$ and subjected to geometric transformations (rotations and reflections) according to fixed recipes. Let (n) denote the identity transformation, $(r180)$ denote a 180° rotation, (fv) denote a vertical flip, and (fh) denote a horizontal flip. The recursion recipes for the upper tile G and lower tile B are defined by:

$$G^{(k)} = \begin{pmatrix} G^{(r180)} & B^{(fv)} & G^{(n)} \\ G^{(fv)} & B^{(n)} & G^{(fh)} \\ B^{(n)} & B^{(fv)} & G^{(n)} \end{pmatrix}, \quad B^{(k)} = \begin{pmatrix} G^{(r180)} & B^{(fv)} & B^{(n)} \\ G^{(fv)} & B^{(n)} & G^{(fh)} \\ B^{(n)} & B^{(fv)} & G^{(n)} \end{pmatrix}$$

where each entry in the matrix represents the type and orientation of the sub-tile at that grid position.

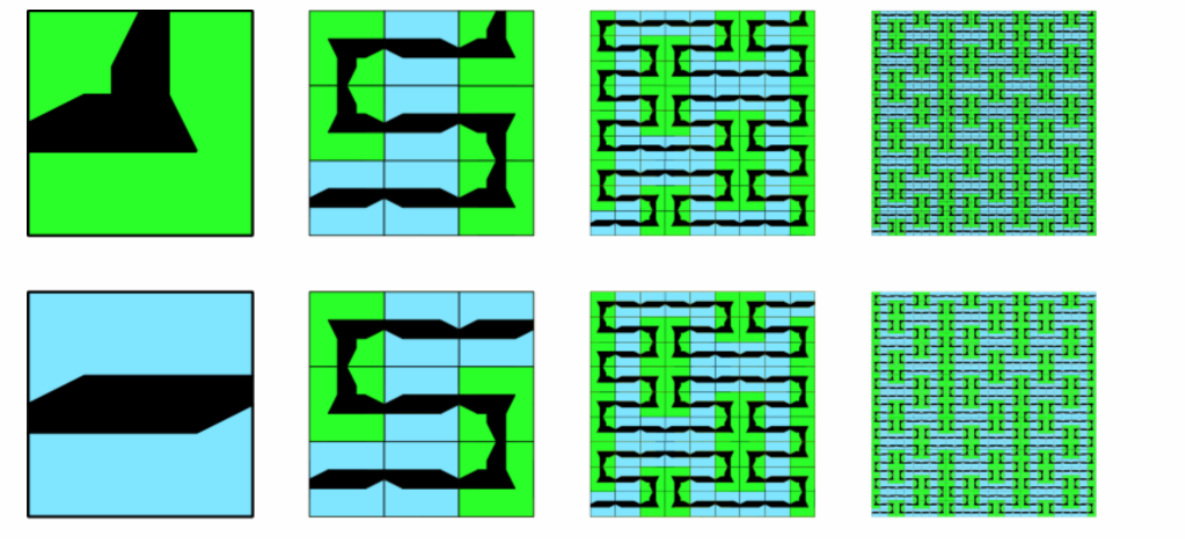


Figure 1: Recursive construction of the Peano curve tiles from Order 0 to Order 3. The first row displays the upper tile (Green, G) and the second row displays the lower tile (Blue, B). The black path represents the curve segment (discrete sink set), which winds to partition the square grid into finer bulk domains (green and blue regions) as the order increases.

On the discrete grid of resolution L, the pixels that lie on the path of the Peano curve are mapped to the sink set V_{sink}, while the green and blue regions form the active bulk domains V_{active}. As the recursion order k increases, the curve winds progressively to fill the space, segmenting the bulk into smaller, isolated pockets.

2 Identity Computation and Verification

The computation of the identity element is directly derived from the mathematical formulation of Dhar's algorithm:

$$E = (\varphi - \varphi^\circ)^\circ$$

where φ represents a sufficiently large configuration, and $(\cdot)^\circ$ denotes the stabilization operation.

In our implementation, we choose $\varphi = 6 \cdot \text{mask}$, where `mask` defines the graph vertices under the Peano curve sink set constraints. The calculation proceeds as follows:

1. Compute the stabilized state $v = \varphi^\circ$ under the open boundary conditions.
2. Form the intermediate state $s = \varphi - v$, and stabilize it to obtain the final identity element $E = s^\circ$.

To validate the result, we numerically verify the fundamental group property of the identity element:

$$(E + E)^\circ = E$$

This checks whether stabilizing the sum of the computed element with itself yields the element itself. The verification successfully confirms that the computed configuration satisfies the group axiom across all tested fractal orders and resolutions.

3 Same Order Identity Configurations at Varying Resolutions

First, I investigate the same order identity configurations of the sandpile group for grid graphs defined by Peano-curve-based sink sets across multiple grid resolutions. A comparison leads to the following sentences:

- As the grid size increases, the identity configurations exhibit similar structures, where finer-scale patterns progressively emerge while the global geometric characteristics are consistently preserved.
- The Peano curve sink divides the graph into distinct subregions where patterns remain locally regular and predictable as resolution increase, whereas highly intricate and unpredictable structures arise exclusively near the sink boundaries.

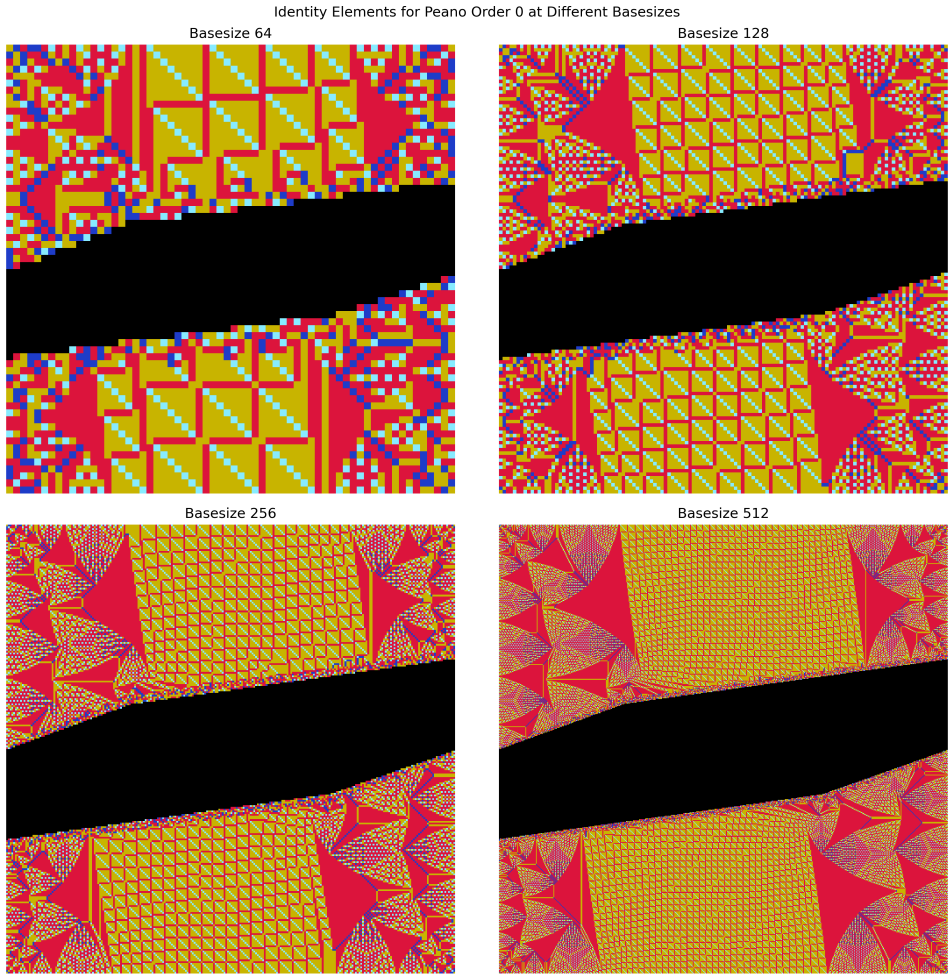


Figure 2: The first order Identity configurations of the sandpile group with Peano-curve-based sink sets at increasing grid resolutions.

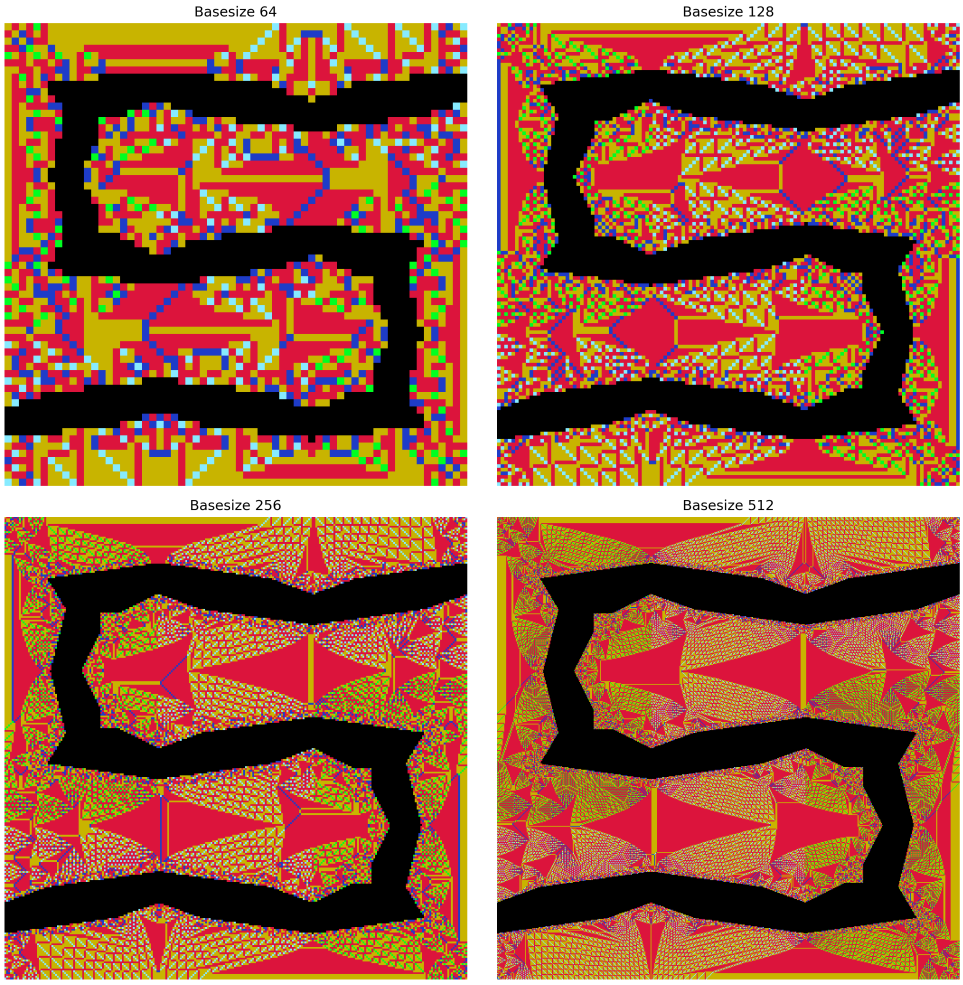


Figure 3: The second order Identity configurations of the sandpile group with Peano-curve-based sink sets at increasing grid resolutions.

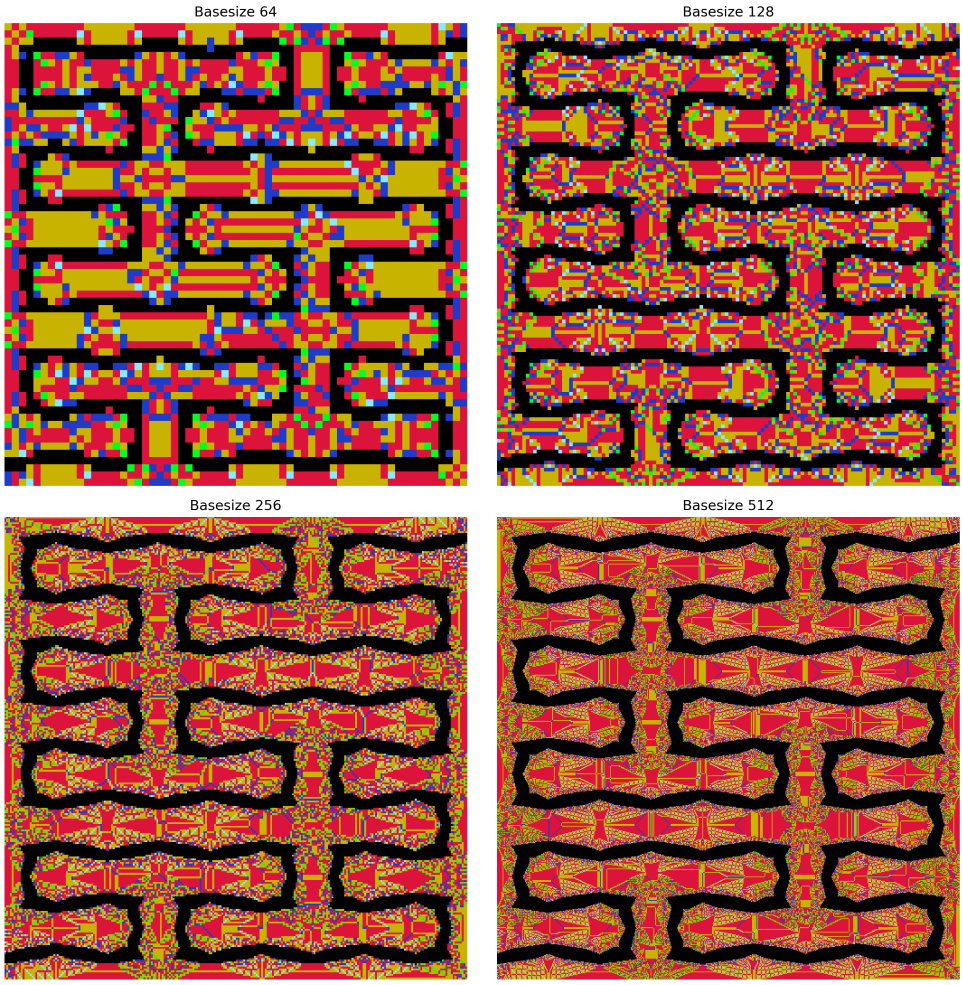


Figure 4: The third order Identity configurations of the sandpile group with Peano-curve-based sink sets at increasing grid resolutions.

4 Identity Configurations of Fixed Resolution (`basesize=512`) at increasing order

At the local scale, the identity configurations of Order i and Order $i+1$ exhibit both continuity and distinct differences.

- Locally, regions far from the sink boundaries retain similar periodic patterns across consecutive orders, reflecting the inherent regularity of the sandpile group identity that is independent of minor sink geometry variations.
- In contrast, significant differences arise near the sink boundaries: compared with Order i, sink of Order $i+1$ breaks the large-scale periodicity of Order i and form localized, discontinuous patterns.

Figure 5: Identity configurations for different Peano curve orders at `basesize=512`.

5 Comparison of Final Height and Odometer (Toppling Function)

Beyond the final height configurations, we further investigate the corresponding toppling functions (odometers) recorded during the stabilization of the identity element.

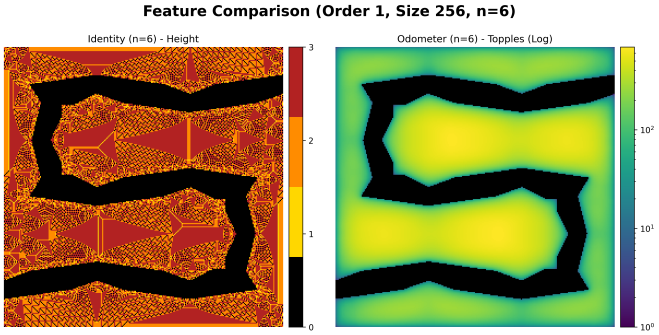
5.1 Mathematical Tracking of the Odometer Function

To track the dynamical potential of the sandpile stabilization, we define the odometer function $u(x)$ which records the cumulative number of toppling events at each vertex x throughout the stabilization process:

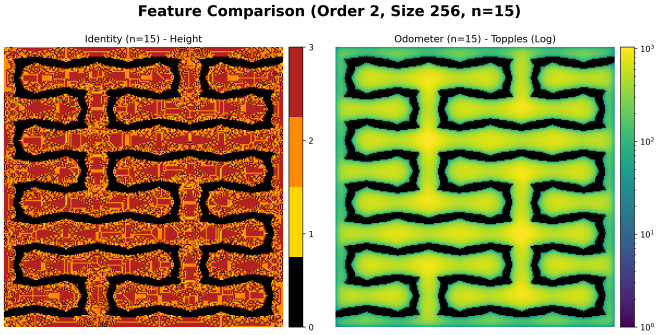
$$u(x) = \sum_{t=1}^{T} \delta_t(x)$$

where t denotes the discrete stabilization step, T is the total number of iterations required to reach equilibrium, and $\delta_t(x) = \lfloor h_t(x)/4 \rfloor$ represents the number of times vertex x topples at step t.

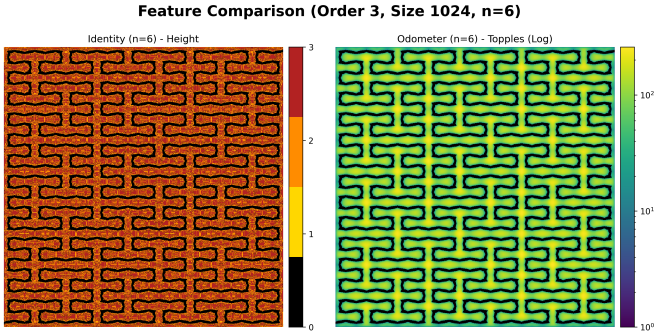
During each iteration, sand is distributed to the four nearest neighbors of any unstable vertex ($h_t(x) \geq 4$). Grains that land on the Peano curve sink set are immediately absorbed and removed from the active system. Accumulating the toppling quotients $\delta_t(x)$ at each step yields the final odometer function $u(x)$, which represents the mathematical potential field governing the sand flow.



(a) (Order 1)



(b) (Order 2)



(c) (Order 3)

Figure 6: Comparison of Identity Height (left half of each plot) and Odometer Function (right half) across different Peano orders. The Odometer consistently reveals a more regular geometric structure than the height maps.

- **Clean Structure and Universal Continuity:** While the final height configuration appears as a complex, seemingly discontinuous pattern, the toppling function reveals a significantly cleaner and more universal structure. In our visualizations, the number of topplings is recorded on a logarithmic scale; darker regions indicate vertices that toppled fewer times. This is particularly insightful for vertices near the sink where avalanches are less frequent. The results illustrate that the fragmented patterns characteristic of the height function are absent in the odometer function, which instead exhibits a much higher degree of spatial continuity.
- **Tolerance and Dynamic Range Analysis:** A primary factor contributing to this universal structure is the difference in dynamic range between the two functions. For each vertex, the value of the toppling function is typically several orders of magnitude higher than the local height (which is capped at 3). This large range allows the odometer to "tolerate" minor local variations between adjacent vertices. In the height function, a difference of just 1 grain represents a 25% shift in value, immediately triggering a sharp color change and forming a discontinuous pattern. In contrast, the odometer function smooths these variations into a continuous geometric gradient.

6 Profile and Growth of the Odometer Function in Uniform Backgrounds

In this section, we transition from studying the *support* of the toppling function (the set of vertices where $u > 0$) to a detailed analysis of its *growth and profile* in uniform initial states. As previously observed, for initial heights $n \geq 4$ on the square grid, almost all non-sink vertices are unstable, causing the support to saturate. Consequently, the shape and magnitude of the odometer function become the primary objects of interest.

6.1 Analytical Methodology

To characterize the odometer function $u(x)$, we performed simulations at two distinct resolutions to observe how fractal complexity influences the results:

- **Low Resolution Case:** Peano Order 1, Basesize 256 ($L = 256$).
- **High Resolution Case:** Peano Order 3, Basesize 1024 ($L = 1024$).

For each case, we examined uniform initial heights $n \in \{4, 6, 8, 10, 12\}$ and measured:

- **Global Maximum ($u_{\max}$):** The peak toppling value and its concentration.
- **Vertical Cross-sectional Profiles:** Slices through the grid center.
- **Level Set Topography:** The evolution of regions $\{x : u(x) \geq k\}$.

6.2 Results and Interpretation: Quantitative Analysis

6.2.1 Global Odometer Profiles and Maximal Concentration

Figures 7 and 8 present the global evolution of the odometer function.

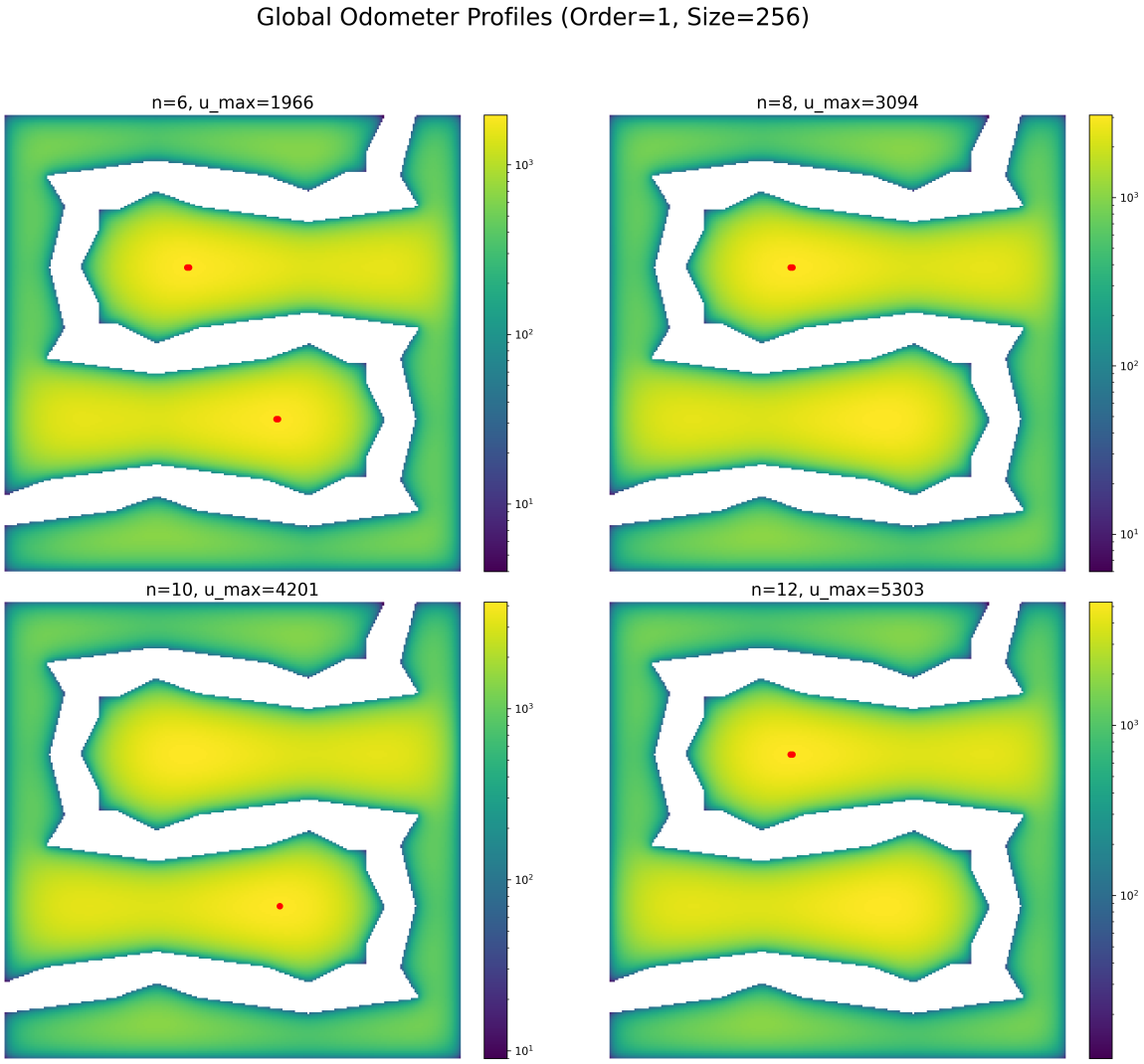


Figure 7: Global odometer profiles (Low Res: Order 1, $L = 256$).

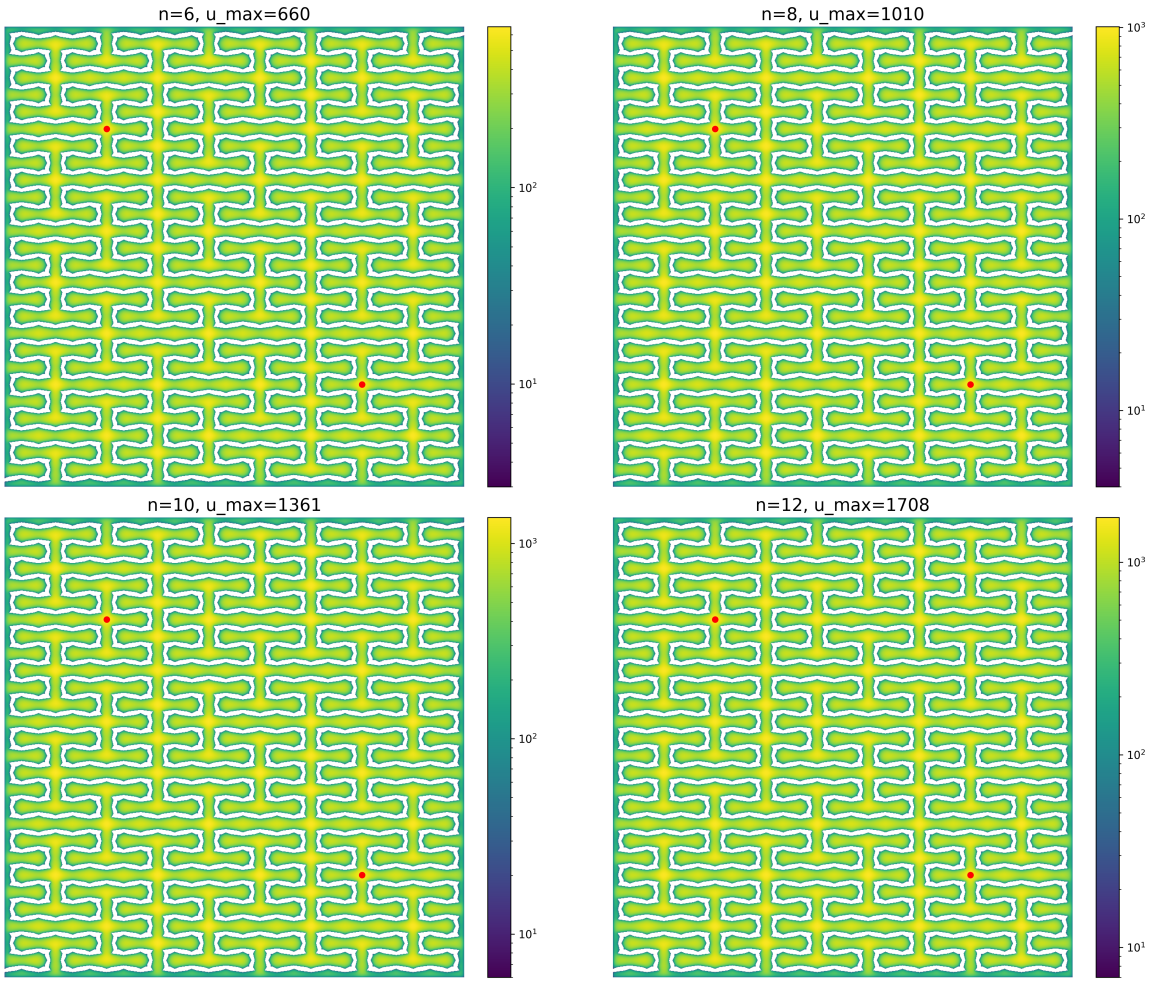


Figure 8: Global odometer profiles (High Res: Order 3, $L = 1024$). Red outlines highlight the highly concentrated maximal sets.

The visualization shows that the odometer attains its global maximum at the geometric centers of the largest connected non-sink components. These maximal sets (highlighted in red) are notably concentrated rather than spread out. .

Technical Note on Numerical Symmetry: Theoretically, given the central symmetry of the Peano sink and the uniform sand injection, the odometer function should exhibit at least two symmetric maximum points. The observed oscillation of $u_{\max}$ between the top-left and bottom-right centroids across different n is a numerical artifact of the discrete grid. This symmetry breaking stems from the nearest-neighbor `resize` algorithm used during fractal generation, which introduces a sub-pixel sampling bias. This effect is most pronounced at low resolutions (*Basesize*=256) and lower recursion orders, where discrete sand-pattern fluctuations are not yet diluted by the scale-averaging observed in the High Resolution case (Order 3).

6.2.2 Cross-sectional Topography and Boundary Effects

To evaluate the spatial uniformity and boundary influence, we examine vertical slices through the grid center (Figures 9 and 10).

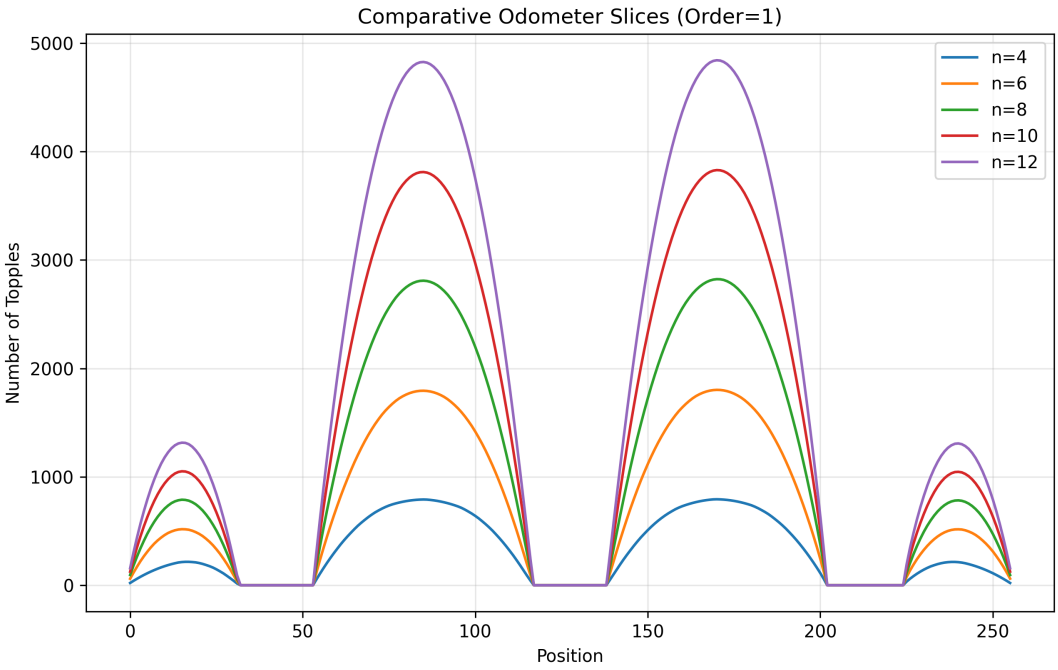


Figure 9: Vertical cross-sections of the odometer function (Low Res: Order 1).

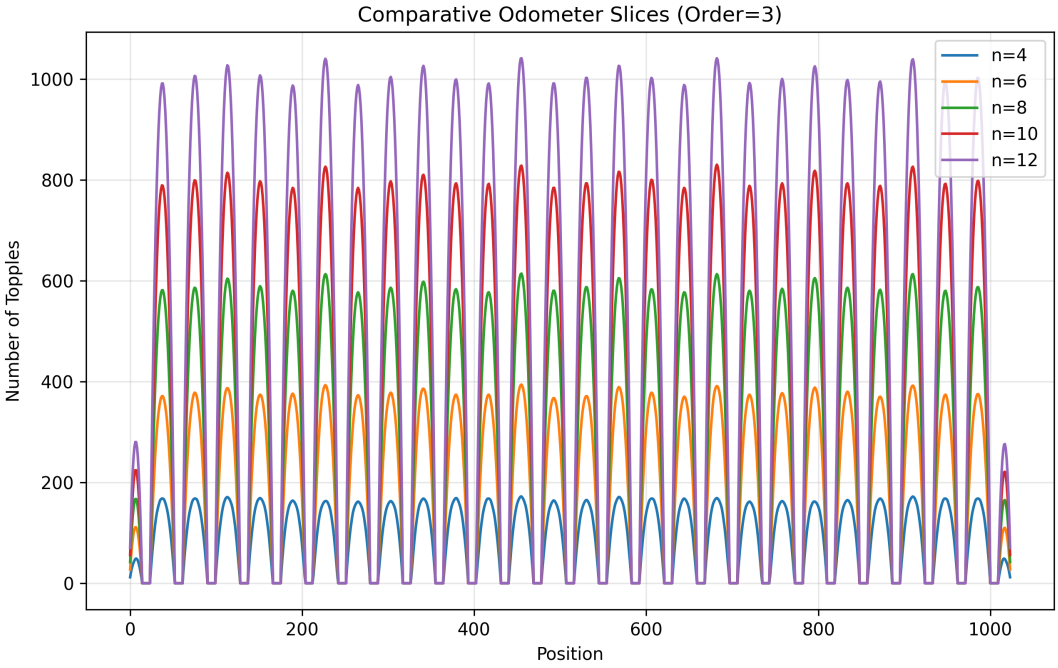


Figure 10: Vertical cross-sections of the odometer function (High Res: Order 3). Zero-value gaps indicate intersections with the Peano sink boundary.

The cross-sections confirm that the toppling function is **strictly non-uniform** in the bulk. The profiles exhibit smooth, parabolic-like arcs in the interior of the regions, which contrast sharply with the behavior near the Peano sink boundaries. Near the boundaries, the odometer drops abruptly to zero. The fractal structure of the Peano sink effectively segments the bulk, preventing any global plateau of uniform toppling and forcing high gradients at every boundary intersection.

6.2.3 Scaling Dynamics and Profile Invariance

The quantitative growth of the odometer relative to the initial sand density n is captured in Figures 11 and 12.

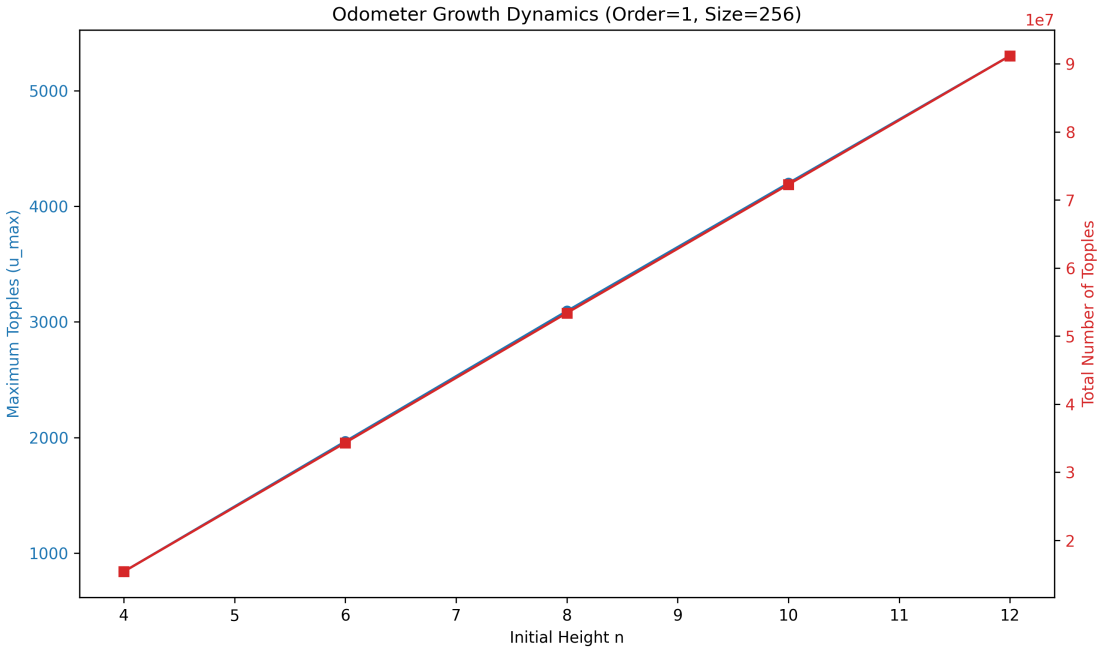


Figure 11: Growth of $u_{\max}$ and total toppling count (Low Res: Order 1).

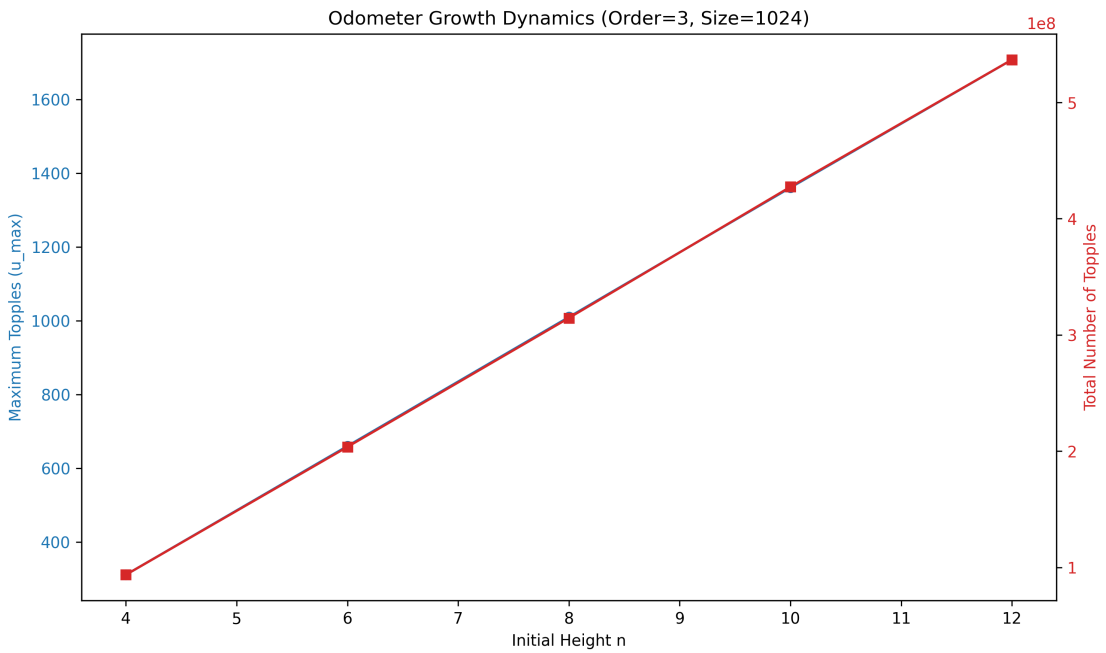
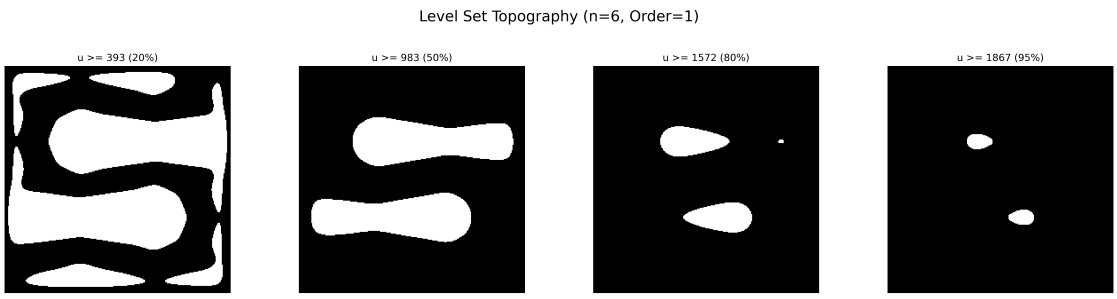


Figure 12: Growth of $u_{\max}$ and total toppling count (High Res: Order 3).

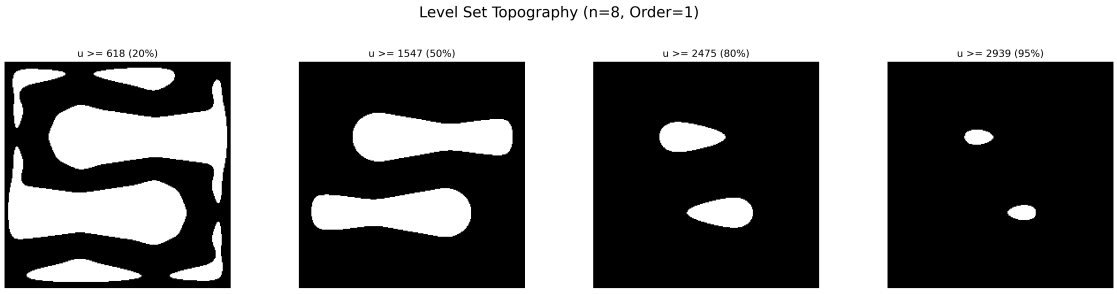
The growth statistics (Figures 11 and 12) reveal a remarkably linear relationship between the initial height n and both the global maximum $u_{\max}$ and the total toppling count for $n \geq 4$. This linear scaling is preserved across resolutions, confirming that the sandpile stabilization behaves as a proportional flux system governed by the underlying graph geometry.

6.2.4 Level Set Topography and Structural Persistence

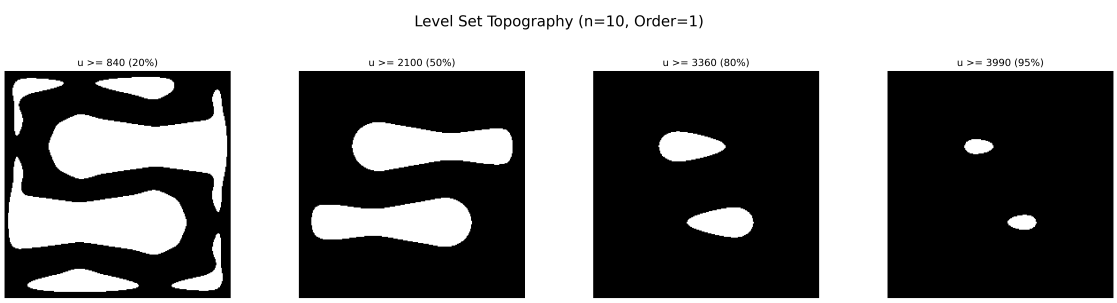
To further visualize the concentration of maxima, we compare the level set evolution (Figures 13 and 14).



(a) Low Res: $n = 6$

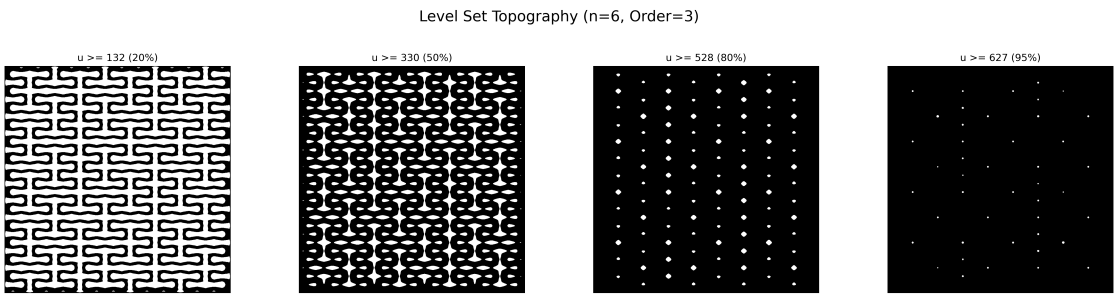


(b) Low Res: $n = 8$

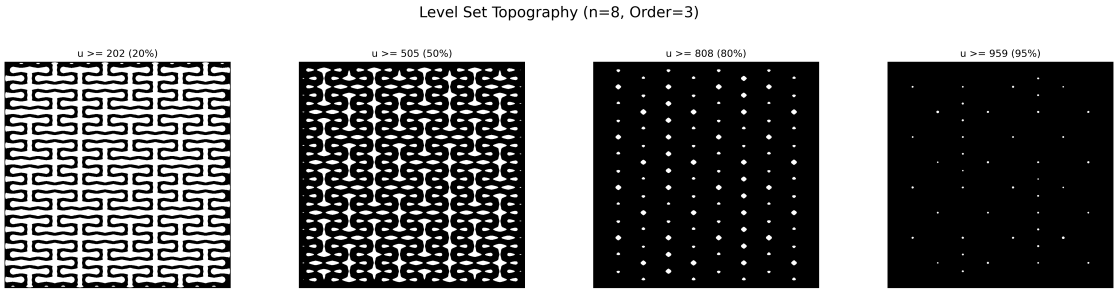


(c) Low Res: $n = 10$

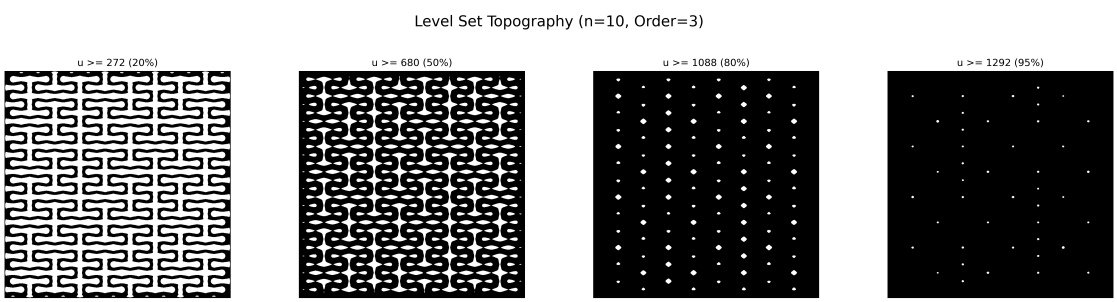
Figure 13: Level set hierarchy for the Low Resolution case (Order 1, $L = 256$).



(a) High Res: $n = 6$



(b) High Res: $n = 8$



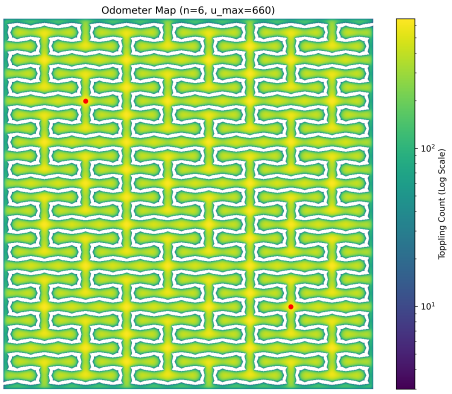
(c) High Res: $n = 10$

Figure 14: Level set hierarchy for the High Resolution case (Order 3, $L = 1024$).

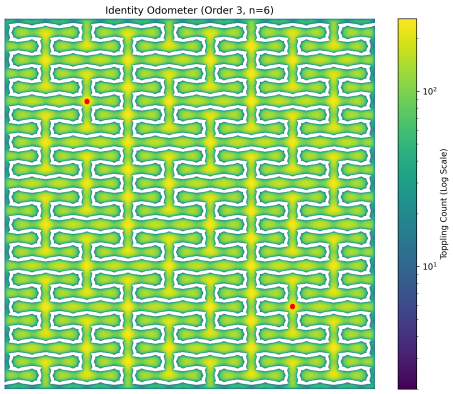
The level set evolution (Figures 13 and 14) provides definitive evidence that the maximal points are concentrated. As the threshold k increases toward $u_{\max}$, the support $\{x : u(x) \geq k\}$ monotonically contracts toward isolated, pixel-scale clusters. This structural persistence across both resolutions and multiple values of n confirms that the odometer's peaks are concentrated.

6.3 Comparison with Identity Odometer

Beyond uniform stabilization, it is instructive to compare the odometer profile u with the toppling function u_{id} generated during the computation of the sandpile group's identity element e.



(a) Uniform Odometer ($n = 6$)



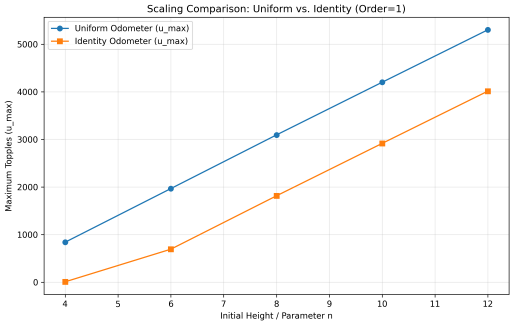
(b) Identity Odometer

Figure 15: Comparison of the uniform stabilization odometer ($n = 6$) and the identity element odometer. Both functions exhibit peaks at identical geometric centroids, though the identity odometer shows a more complex, multi-scale internal structure.

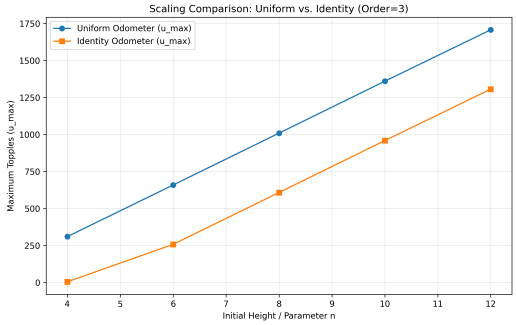
The identity odometer (Figure 15) effectively "explores" the global capacity of the graph. While its absolute magnitude and total toppling count are lower than those in the uniform case, the location and geometry of the maximal set remain remarkably consistent with the uniform case. This suggests that the Peano sink's geometry acts as a universal potential barrier that dictates the flow of sand regardless of the specific stabilization process.

6.3.1 Quantitative Profile Comparison

To further analyze the relationship between the uniform and identity cases, we examine the cross-sectional profiles (Figure 16).



(a) Order 1 Comparison



(b) Order 3 Comparison

Figure 16: Growth comparison of $u_{\max}$ between the uniform case and the identity element calculation. Both processes follow similar scaling laws, with the identity odometer maintaining a lower absolute magnitude due to its reduced initial sand density.

The comparison in Figure 16 reveals several critical insights into the fractal sandpile dynamics:

- **Scaling Universality:** For $n \geq 6$, the growth curves for both the uniform stabilization and the identity element calculation exhibit nearly identical slopes. This parallel scaling suggests that the

Peano sink geometry dictates once a certain threshold of sand density is reached, the additional work required to stabilize the system follows a predictable geometric progression, independent of the initial configuration's algebraic nature.

- **Identity Regime Transition ($n = 4 \to 6$):** A notable phenomenon is the lower slope observed in the identity odometer between $n = 4$ and $n = 6$. For $n = 4$, the identity verification frequently fails, indicating that the background height is insufficient to support the canonical identity element E of the sandpile group. Consequently, the odometer at $n = 4$ represents a "premature" stabilization process. As n increases to 6, the system reaches a critical density required to form the stable identity pattern, leading to a non-linear adjustment in the total toppling count before settling into the universal scaling regime.
- **Magnitude Difference:** The identity odometer maintains a consistently lower magnitude ($u_{\max}$) than the uniform case. This is because the identity's initial state is derived by subtracting a stabilized configuration from the uniform background. Since this initial state contains fewer total grains than the uniform n injection, the total effort required to reach equilibrium is reduced, while still preserving the underlying geometric structure.

7 Macroscopic Limit: Discrete Laplacian and Smoothing Analysis

To analyze the macroscopic limit of the sandpile configurations and filter out high-frequency microscopic fluctuations, we compute the discrete Laplacian of the final height configuration h and apply a spatial smoothing filter.

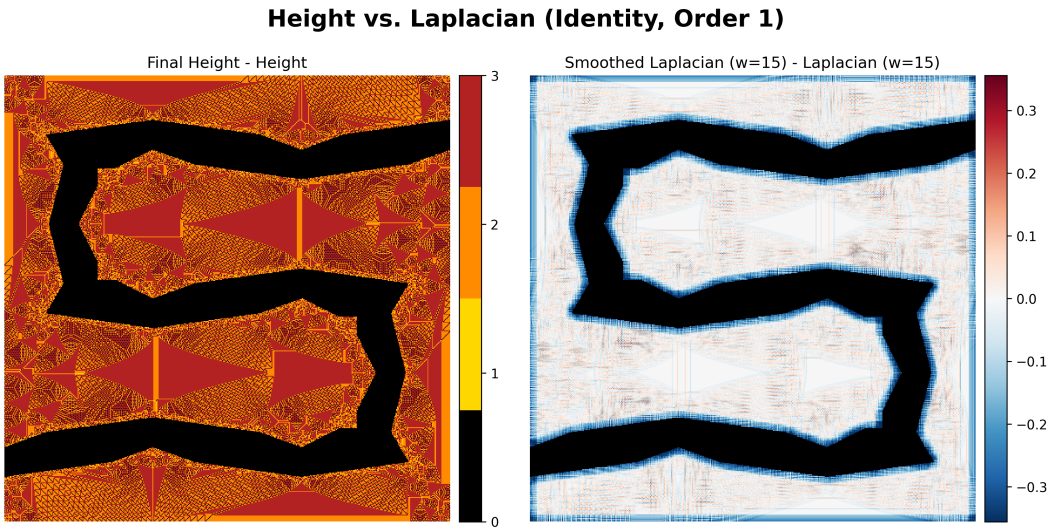
For a discrete height configuration $h(x, y)$ on a 2D square grid, the discrete Laplacian $\Delta h(x, y)$ at an active vertex (x, y) is defined by:

$$\Delta h(x, y) = h(x + 1, y) + h(x - 1, y) + h(x, y + 1) + h(x, y - 1) - 4h(x, y)$$

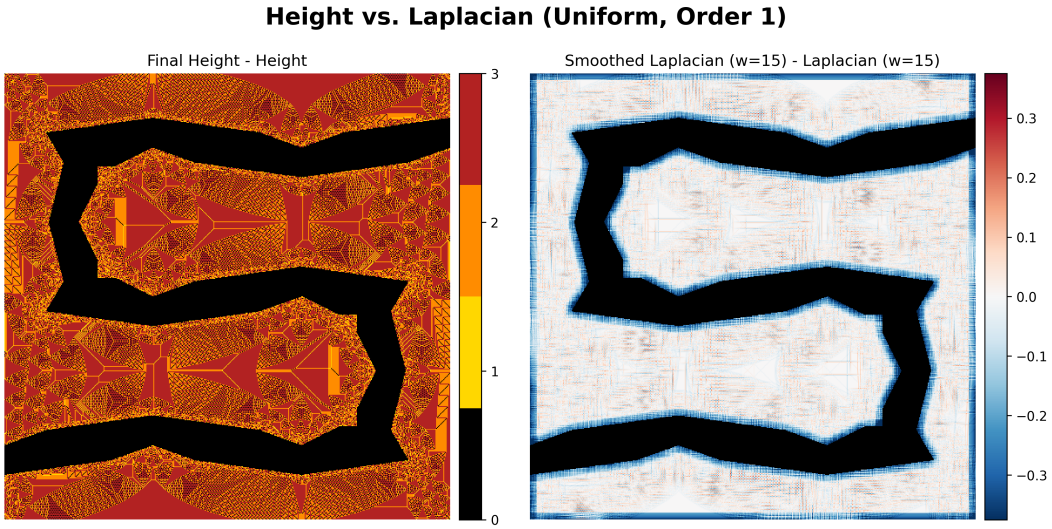
To smooth the resulting Laplacian field and eliminate pixel-level noise, we apply a local moving average filter over a square window of size $w \times w$ (where $w = 15$). Because the grid contains boundary vertices near the complex Peano curve sink set (where the values are undefined or treated as zero), standard convolution would lead to boundary bleeding. To resolve this, we implement a normalized convolution:

$$\bar{\Delta} h(x, y) = \frac{\sum_{i,j \in [-7,7]} \Delta h(x + i, y + j) \cdot M(x + i, y + j)}{\sum_{i,j \in [-7,7]} M(x + i, y + j)}$$

where $M(u, v)$ is a binary mask defined as 1 for active bulk vertices and 0 for sink vertices.



(a) Identity Configuration



(b) Uniform Stabilization

Figure 17: Comparison of Final Height (left) and Smoothed Laplacian (right) for Peano Order 1 ($L = 512$, $n = 6$). The smoothed Laplacian filters out high-frequency discrete noise while preserving the global macroscopic structure.

7.1 Visual Observations and Phenomenon Description

- **Erasure of Microscopic Noise:** The original height configurations (left halves) are highly discrete and pixelated, characterized by sharp integer height values (0 to 3) that form high-frequency, complex grid textures. In contrast, the smoothed Laplacian maps (right halves) filter out these pixel-level fluctuations, resulting in a smooth, continuous gradient field.
- **Preservation of Macroscopic Structures:** Despite the blur of microscopic details, the large-scale geometric features are clearly preserved. The prominent triangular shapes remain visible while the smooth decay toward the Peano curves and outer boundaries is clearly outlined.

7.2 Physical Implications

- **Low-Pass Filtering and Scale Separation:** The combination of the discrete Laplacian and spatial smoothing acts as a spatial low-pass filter. It successfully separates the local microscopic fluctuations from the global macroscopic field.

8 Conclusion

In this project, we have systematically investigated the Abelian Sandpile Model (ASM) on grid graphs bounded by recursive Peano fractal sink sets. By tracing the system's behavior across resolutions and fractal orders, the report establishes a central physical insight: **the transition from microscopic discrete chaos to macroscopic geometric determinism**.

- **Microscopic Fragmentation vs. Macroscopic Order:** At the grid level, the sandpile configurations (final heights) are highly discontinuous and sensitive to boundary variations. As the Peano curve order increases, it acts as a fractal scissor, cutting the bulk into tiny pockets and breaking local periodic patterns. However, at the macroscopic level, the odometer function (toppling count) filters out this discrete noise, revealing smooth parabolic potential fields with concentrated peaks at geometric centroids.
- **Algebraic and Physical Convergence:** A key finding of this study is the convergence of two conceptually distinct processes: the algebraic Identity element (Dhar's formula) and the physical Uniform stabilization. Although their microscopic heights and local dynamics are entirely different, their macroscopic odometer profiles and smoothed Laplacians are almost identical in shape, peak locations, and linear growth scaling ($n \geq 6$). This indicates that under high sand density, the algebraic group structures and physical relaxation processes are both dominated by the same geometric potential barriers.
- **Macroscopic Limit and Potential Theory:** The smoothed Laplacian filters out high-frequency discrete noise while preserving the global macroscopic structure. The macroscopic limit of the discrete sandpile is fundamentally a continuous potential field governed entirely by the boundary geometry of the Peano curve.

In conclusion, the project demonstrates that while the microscopic state of a sandpile on a fractal boundary is chaotic and fragmented, its macroscopic physics is clean, predictable, and strictly determined by the geometry of the domain.